



PHYS121 Integrated Science-Physics

W6T3 The 1st law of Thermodynamics

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
 - [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
 - [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)
- And others specified when needed.



Learning Outcomes

- Apply conservation of energy (the first law of thermodynamics).
- Describe phase transition processes and mole specific heat.
- Explain the adiabatic process and the Van der Waals equation of state.

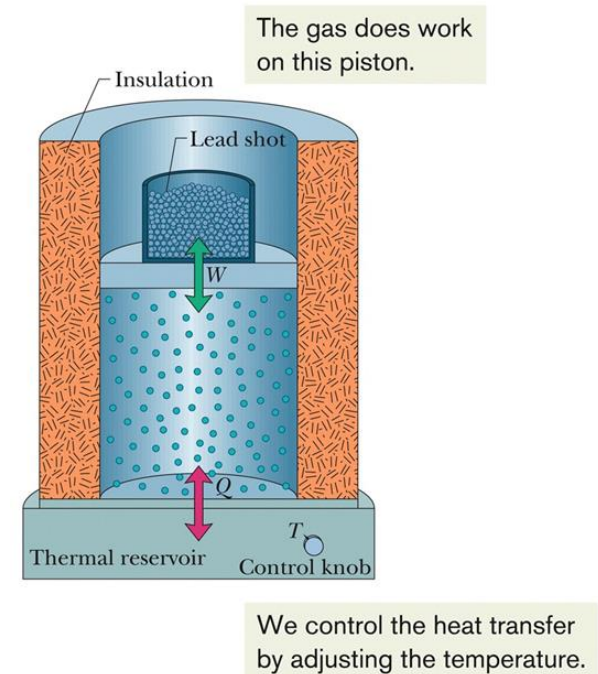
The First Law of Thermodynamics

Heat and Work (revisit)

A gas may exchange energy with its surroundings through work. The amount of work W done by a gas as it expands or contracts from an initial volume V_i to a final volume V_f is given by

$$W = \int dW = \int_{V_i}^{V_f} p \, dV.$$

The integration is necessary because the pressure p may vary during the volume change.



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A gas confined to a cylinder with a movable piston.

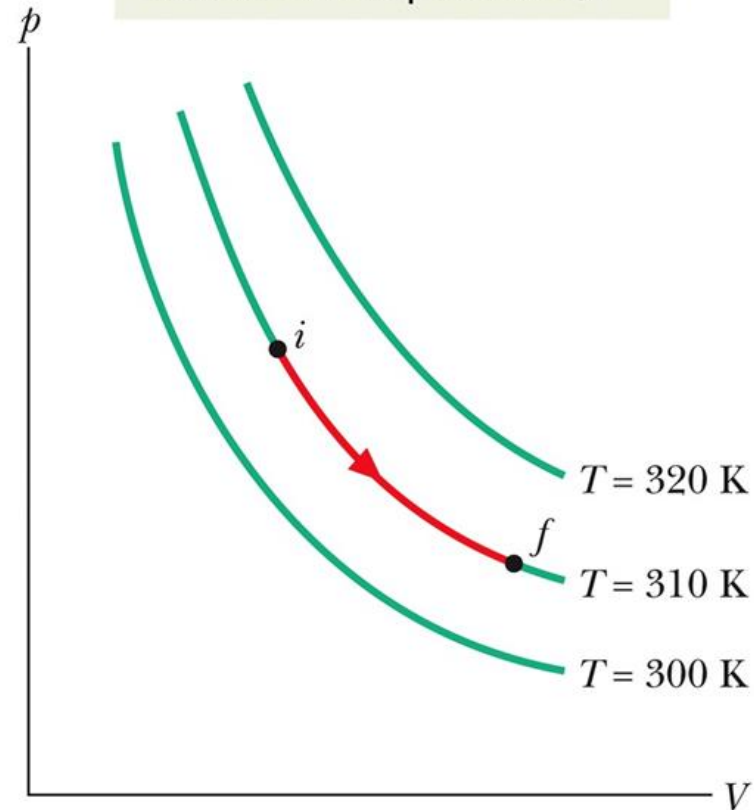
(revisit)

The work done by an ideal gas during an isothermal (constant-temperature) change from volume V_i to volume V_f is

$$W = nRT \ln \frac{V_f}{V_i}$$

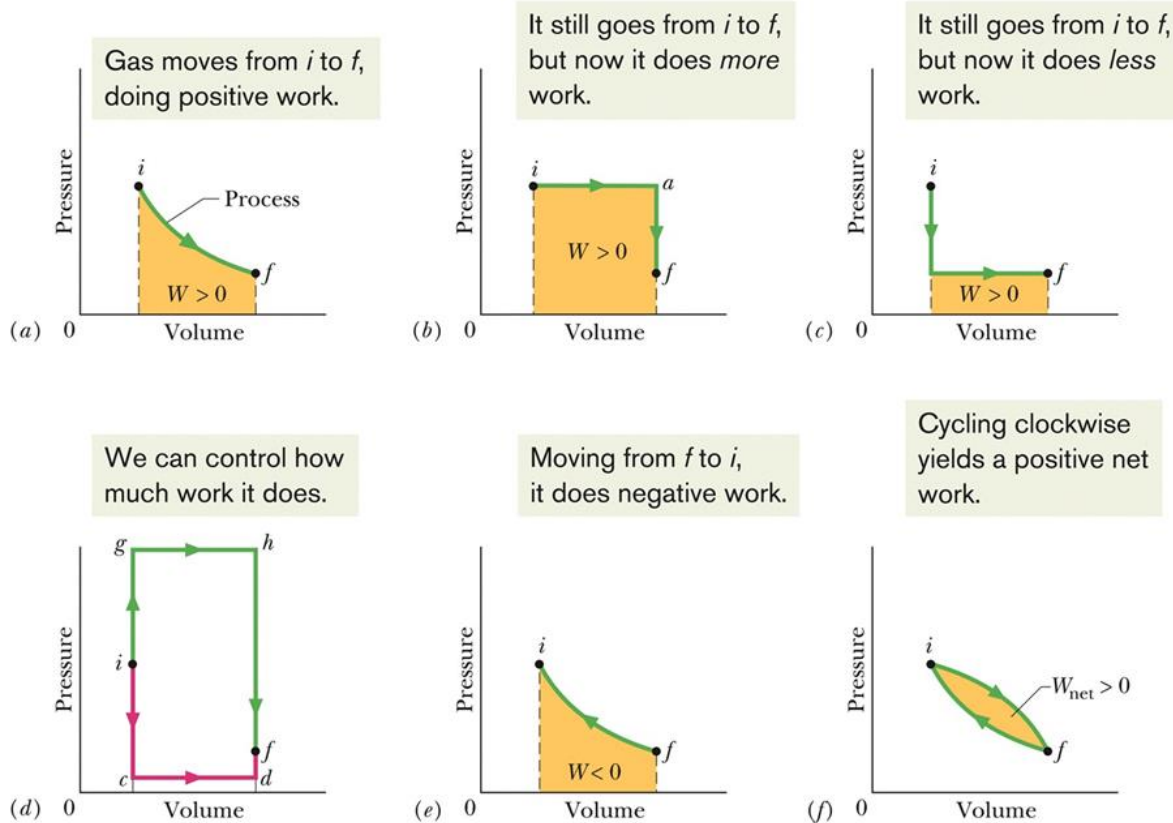
Three isotherms on a p - V diagram. The path shown along the middle isotherm represents an isothermal expansion of a gas from an initial state i to a final state f . The path from f to i along the isotherm would represent the reverse process — that is, an isothermal compression.

The expansion is along an isotherm (the gas has constant temperature).

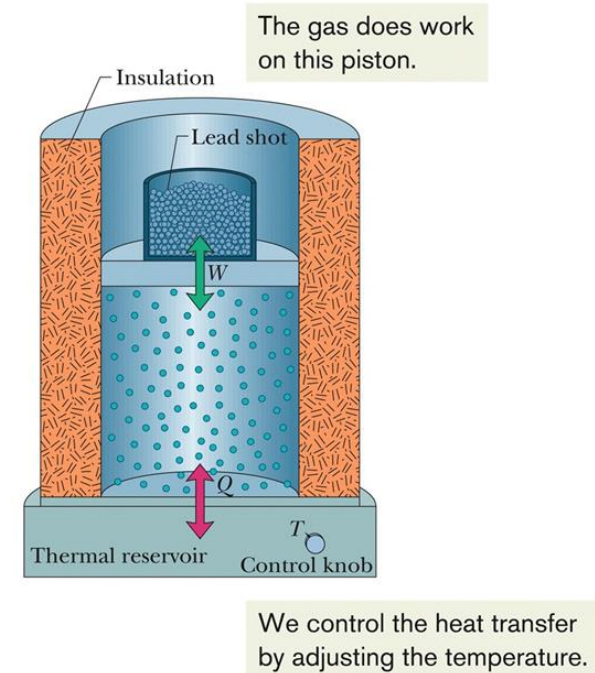


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Heat and Work



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A gas confined to a cylinder with a movable piston.

The First Law of Thermodynamics

The principle of conservation of energy for a thermodynamic process is expressed in the first law of thermodynamics, which may assume either of the forms:

$$\Delta E_{\text{int}} = E_{\text{int}, f} - E_{\text{int}, i} = Q - W \quad (\text{first law}).$$

Or, if the thermodynamic system undergoes only a differential change, we can write the first law as:

$$dE_{\text{int}} = dQ - dW \quad (\text{first law}).$$

The internal energy E_{int} of a system tends to increase if energy is added as heat Q and tends to decrease if energy is lost as work W done by the system.

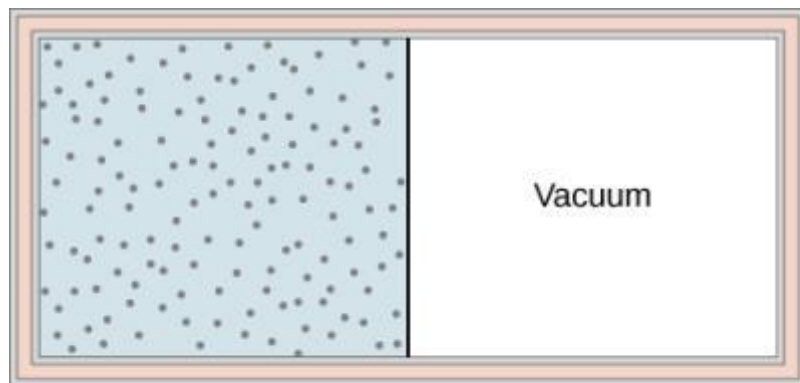
Table 18-5 The First Law of Thermodynamics: Four Special Cases

The Law : $\Delta E_{\text{int}} = Q - W$ (Equations. 18-26)

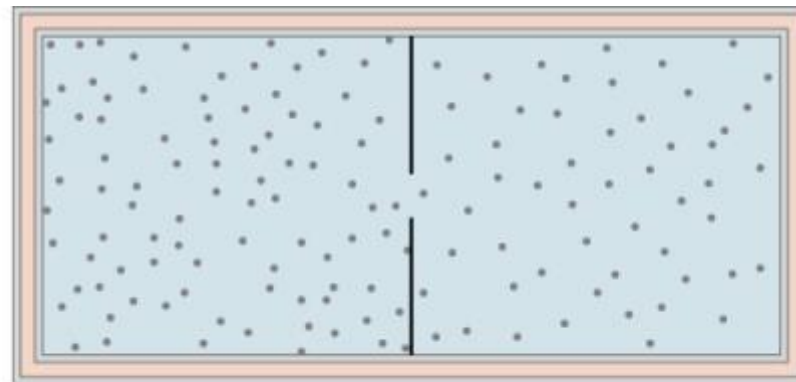
Process	Restriction	Consequence
Adiabatic	$Q = 0$	$\Delta E_{\text{int}} = -W$
Constant volume	$W = 0$	$\Delta E_{\text{int}} = Q$
Closed cycle	$\Delta E_{\text{int}} = 0$	$Q = W$
Free expansion	$Q = W = 0$	$\Delta E_{\text{int}} = 0$

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FIGURE 3.13



Initial equilibrium state



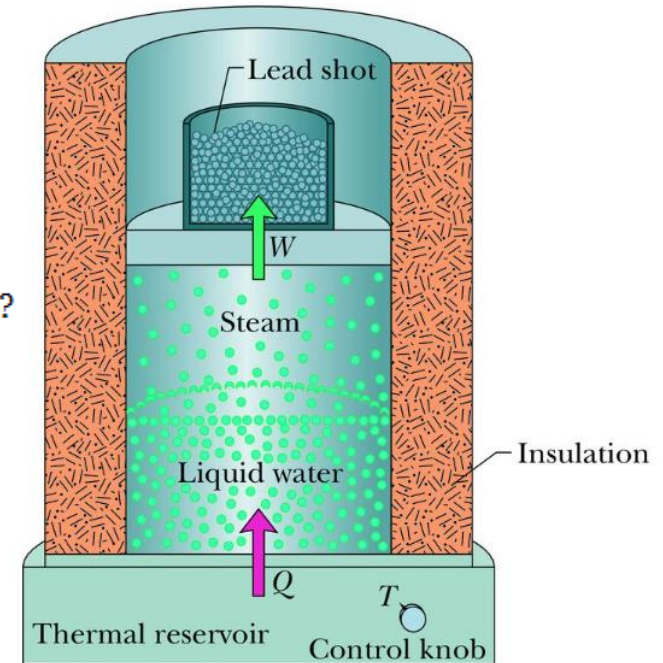
Final equilibrium state

The gas in the left chamber expands freely into the right chamber when the membrane is punctured.

Sample Problem 18.05 First law of thermodynamics: work, heat, internal energy change

Let 1.00 kg of liquid water at 100°C be converted to steam at 100°C by boiling at standard atmospheric pressure (which is 1.00 atm or $1.01 \times 10^5 \text{ Pa}$) in the arrangement of Fig. 18-17. The volume of that water changes from an initial value of $1.00 \times 10^{-3} \text{ m}^3$ as a liquid to 1.671 m^3 as steam.

- (a) How much work is done by the system during this process?
- (b) How much energy is transferred as heat during the process?
- (c) What is the change in the system's internal energy during the process?



Revisit

Molar Specific Heat at Constant Volume

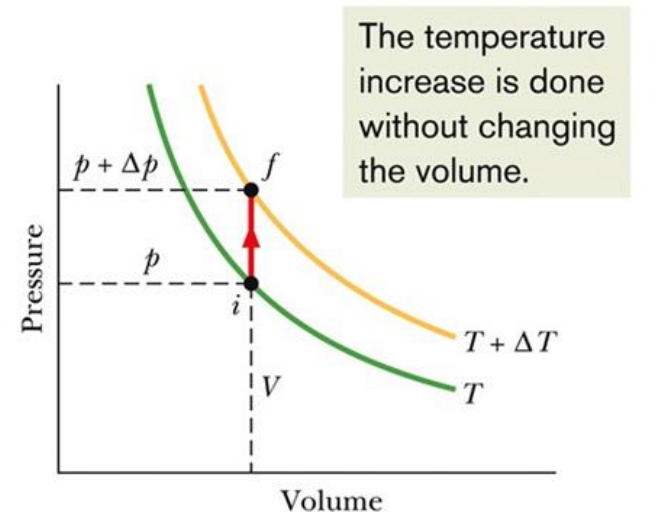
The internal energy E_{int} of an ideal gas (**monatomic**) is a function of the gas temperature only; it does not depend on any other variable.

$$E_{\text{int}} = \frac{3}{2} NkT = \frac{3}{2} nRT$$

The **molar specific heat** of a gas at **constant volume** is defined to be

$$Q = nC_V\Delta T$$

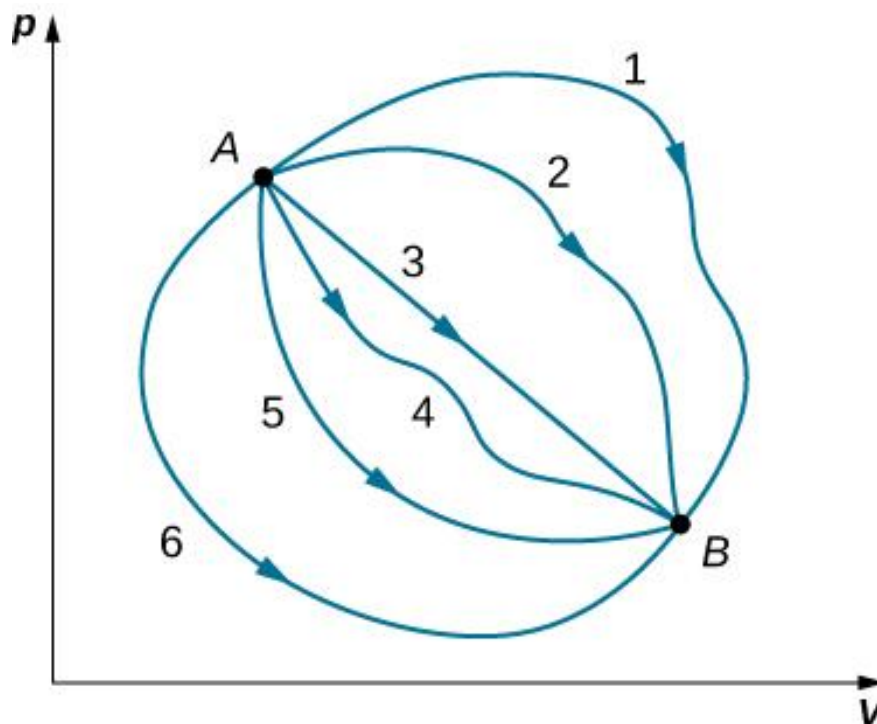
A change in the internal energy E_{int} of a confined ideal gas depends on only the change in the temperature, not on what type of process produces the change.



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$$\Delta E_{\text{int}} = nC_V\Delta T$$

FIGURE 3.7



Different thermodynamic paths taken by a system in going from state A to state B . For all transitions, the change in the internal energy of the system $\Delta E_{\text{int}} = Q - W$ is the same.

Molar Specific Heat at Constant Pressure

The **molar specific heat** of a gas at **constant pressure** is defined to be

$$C_p = \frac{Q}{n\Delta T}$$

where C_p can also be defined as

$$C_p = C_V + R.$$

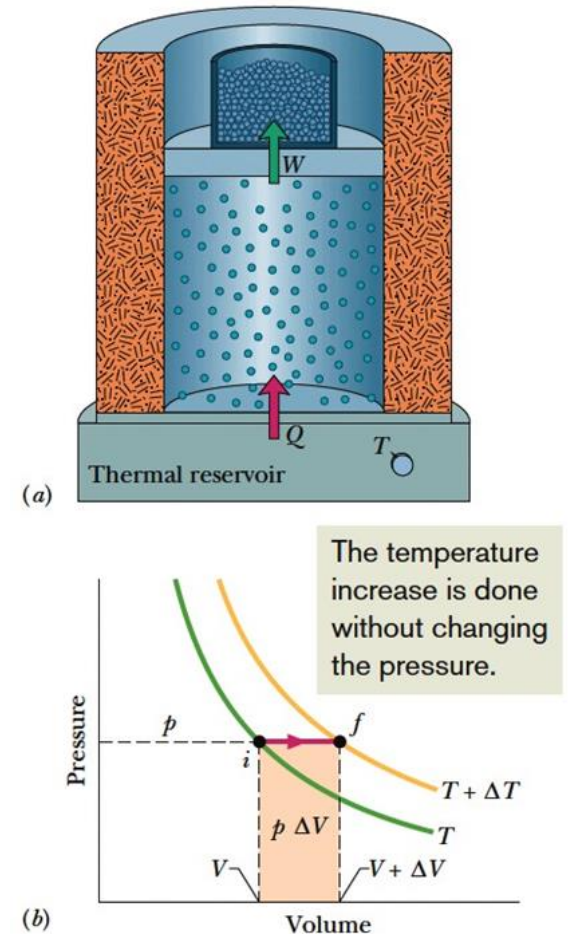
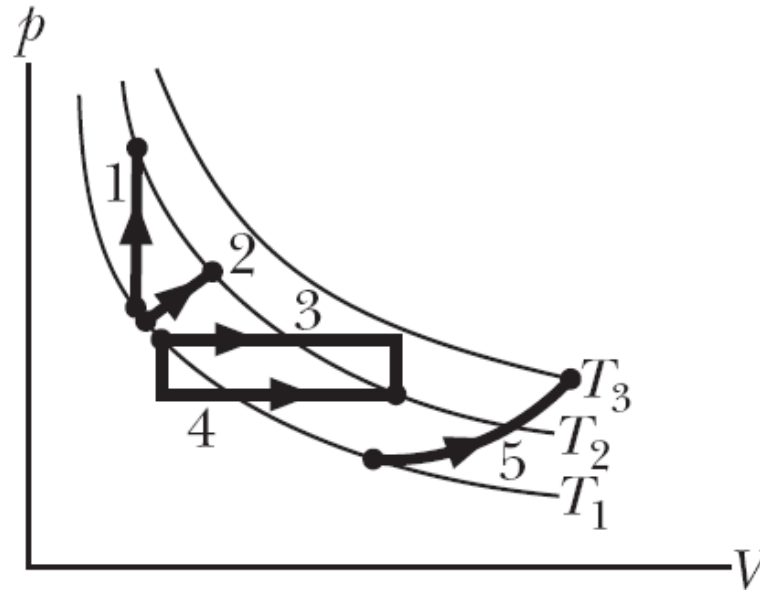


Figure 19-11 (a) The temperature of an ideal gas is raised from T to $T + \Delta T$ in a constant-pressure process. Heat is added and work is done in lifting the loaded piston. (b) The process on a p - V diagram. The work $p \Delta V$ is given by the shaded area.

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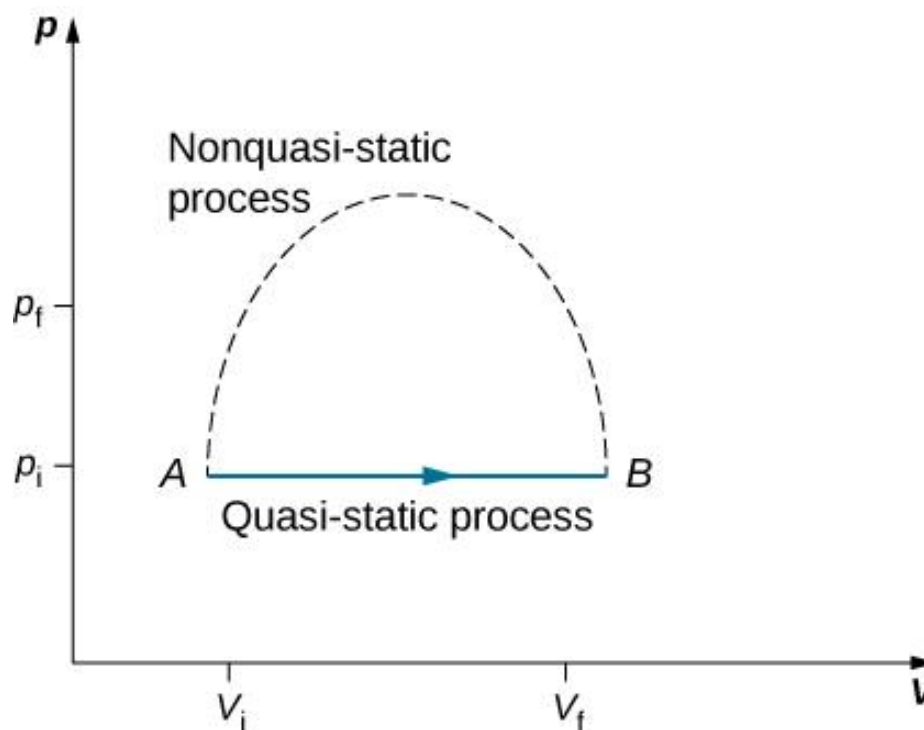
Checkpoint 4

The figure here shows five paths traversed by a gas on a p - V diagram. Rank the paths according to the change in internal energy of the gas, greatest first.



Answer: 5 and then—4,3,2,1—with same int. energy

FIGURE 3.8



Quasi-static and non-quasi-static processes between states A and B of a gas. In a quasi-static process, the path of the process between A and B can be drawn in a state diagram since all the states that the system goes through are known. In a non-quasi-static process, the states between A and B are not known, and hence no path can be drawn. It may follow the dashed line as shown in the figure or take a very different path.

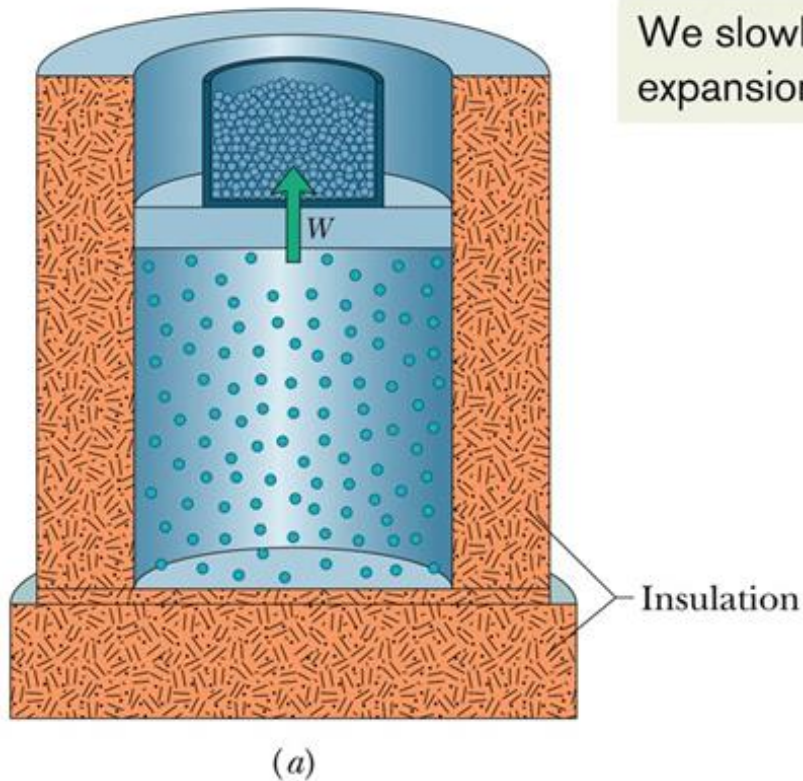
Adiabatic Expansion of an Ideal Gas

When an ideal gas undergoes a slow adiabatic volume change (a change for which $Q = 0$),

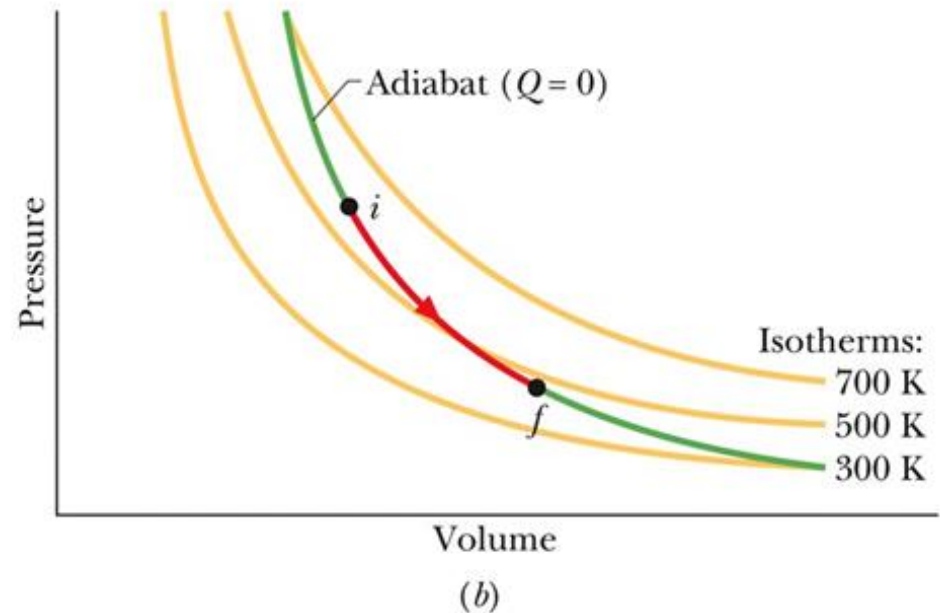
$$pV^\gamma = \text{a constant} \quad (\text{adiabatic process}),$$

in which $\gamma \left(= \frac{c_p}{c_v} \right)$ is the ratio of molar specific heats for the gas.

- (a) The volume of an ideal gas is increased by removing mass from the piston. The process is adiabatic ($Q = 0$).
- (b) The process proceeds from i to f along an adiabat on a p - V diagram.



We slowly remove lead shot, allowing an expansion without any heat transfer.



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Figure 19-15 (a) The volume of an ideal gas is increased by removing mass from the piston. The process is adiabatic ($Q = 0$). (b) The process proceeds from i to f along an adiabat on a p - V diagram.

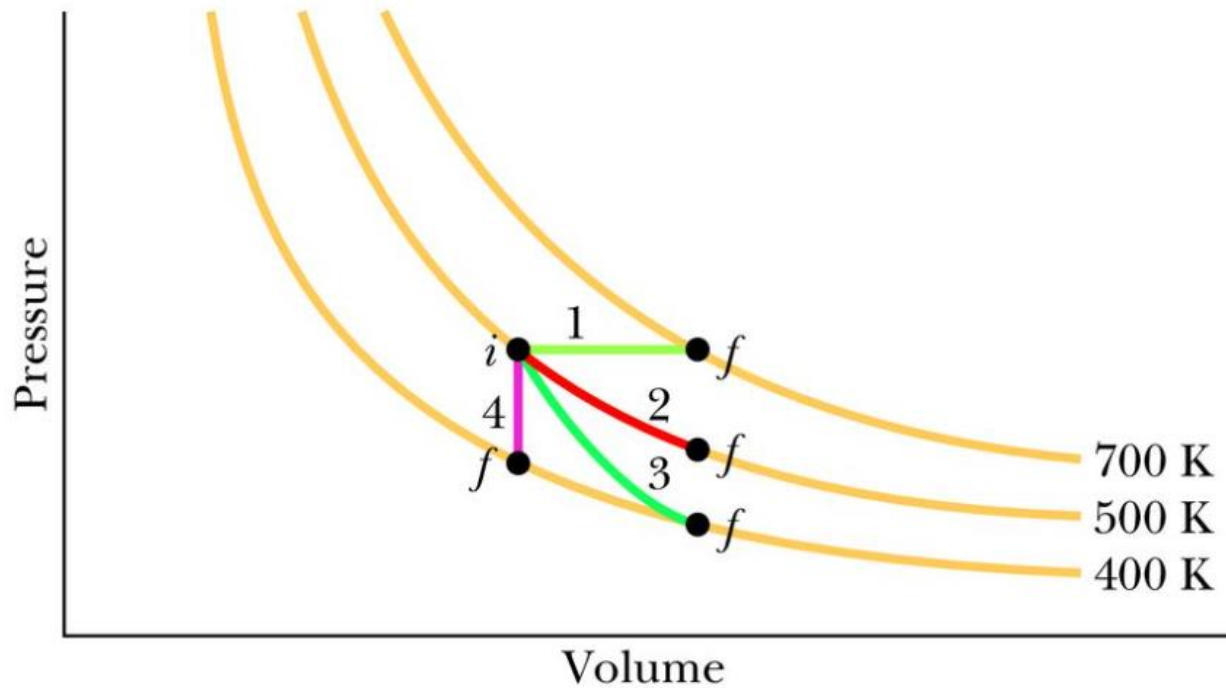
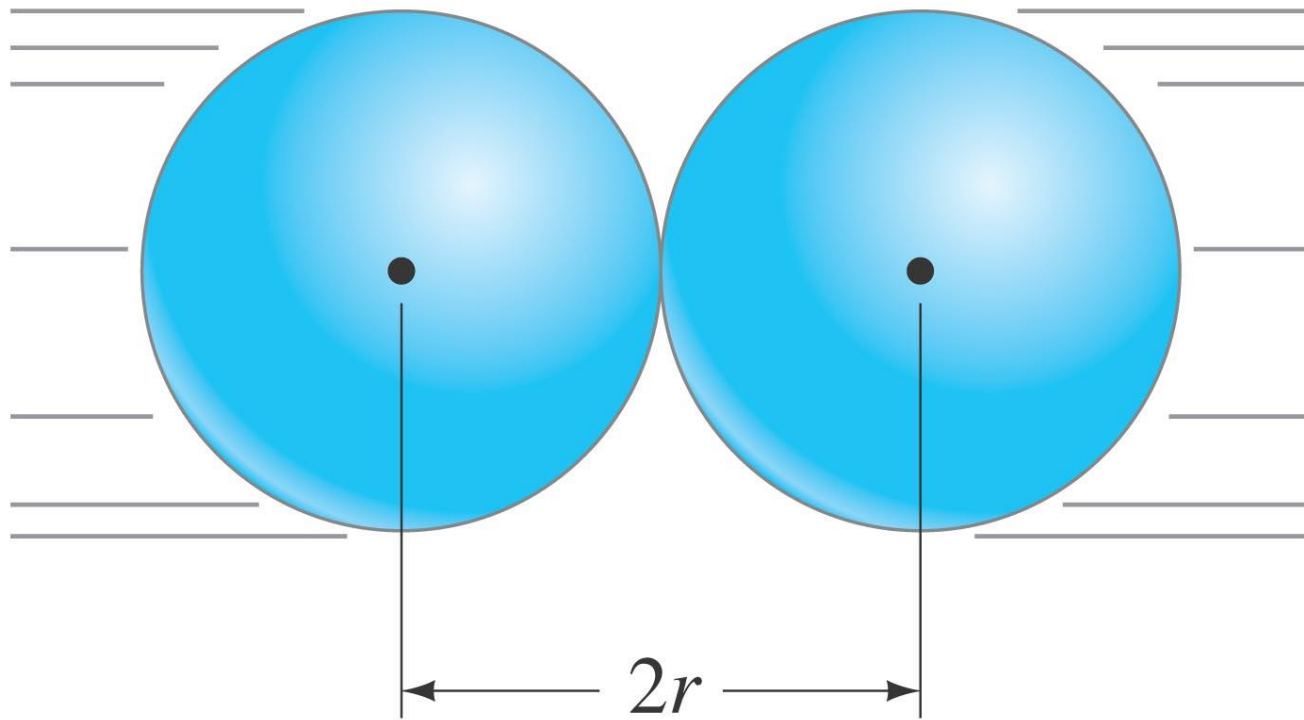


Figure 19-16 A p - V diagram representing four special processes for an ideal monatomic gas.

Rank paths 1, 2, and 3 in Fig. 19-16 according to the energy transfer to the gas as heat, greatest first.

Van der Waals Equation of State

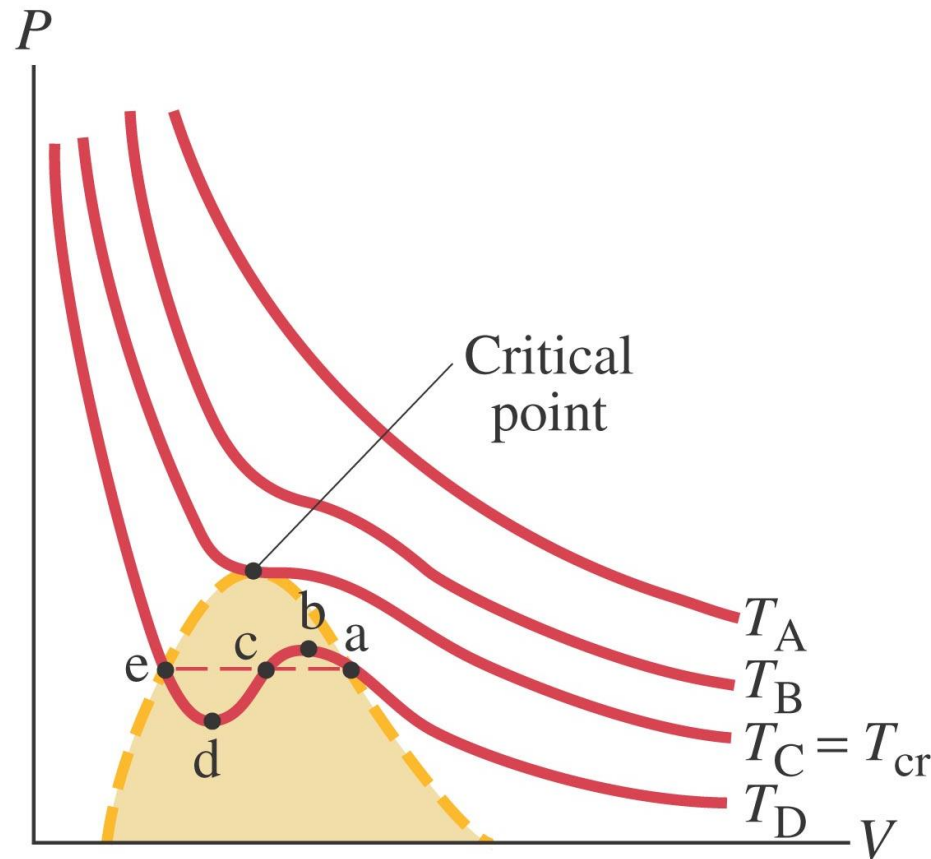
To get a more realistic model of a gas, we include the finite size of the molecules and the range of the intermolecular force beyond the size of the molecule.



We assume that some fraction b of the volume is unavailable due to the finite size of the molecules. We also expect that the pressure will be reduced by a factor proportional to the square of the density, due to interactions near the walls. This gives the **Van der Waals equation of state**; the constants a and b are found experimentally for each gas:

$$\left(P + \frac{a}{(V/n)^2} \right) \left(\frac{V}{n} - b \right) = RT.$$

The PV diagram for a Van der Waals gas fits most experimental data quite well.





Learning Outcomes

- ✓ Apply conservation of energy (the first law of thermodynamics).
- ✓ Describe phase transition processes and mole specific heat.
- ✓ Explain the adiabatic process and the Van der Waals equation of state.

$$\Delta E_{\text{int}} = E_{\text{int}, f} - E_{\text{int}, i} = Q - W \quad (\text{first law}).$$

$$C_V = \frac{Q}{n\Delta T} = \frac{\Delta E_{\text{int}}}{n\Delta T} \quad C_p = C_V + R.$$

$$pV^\gamma = \text{a constant} \quad (\text{adiabatic process}), \quad \gamma \left(= \frac{C_p}{C_V} \right)$$

$$\left(P + \frac{a}{(V/n)^2} \right) \left(\frac{V}{n} - b \right) = RT.$$

CHAPTER REVIEW AND EXAMPLES, DEMOS

<https://openstax.org/books/university-physics-volume-2/pages/3-summary>

A system is called an isolated and **closed system** if it is completely separated from its environment—for example, a gas that is surrounded by immovable and thermally insulating walls. In reality, a closed system does not exist unless the entire universe is treated as the system, or it is used as a model for an actual system that has minimal interactions with its environment.

the **equation of state** of the system and is written symbolically *for a closed system* as $f(p,V,T)=0$, where V , p , and T are the volume, pressure, and temperature of the system at a given condition.

We can separate these quantities into two generic categories. The quantity associated with an amount of matter is an **extensive variable**, such as the volume and the number of moles. The other properties of a system are **intensive variables**, such as the pressure and temperature. An extensive variable doubles its value if the amount of matter in the system doubles, provided all the intensive variables remain the same. For example, the volume or total energy of the system doubles if we double the amount of matter in the system while holding the temperature and pressure of the system unchanged.

This integral is only meaningful for a **quasi-static process**, which means a process that takes place in infinitesimally small steps, keeping the system at thermal equilibrium. ... Only then does a well-defined mathematical relationship (the equation of state) exist between the pressure and volume. This relationship can be plotted on a pV diagram of pressure versus volume, where the curve is the change of state. We can approximate **such a process as one that occurs slowly, through a series of equilibrium states**.

... a very important property of **thermodynamic work**: It is *path dependent*.

The **internal energy** E_{int} of a thermodynamic system is, by definition, the sum of the **mechanical energies** of all the molecules or entities in the system.

Thermodynamic Sign Conventions for Heat and Work	
Process	Convention
Heat added to system	$Q > 0$
Heat removed from system	$Q < 0$
Work done by system	$W > 0$
Work done on system	$W < 0$

Table 3.1

Functions such as internal energy and potential energy are known as *state functions* because their values depend solely on the state of the system.

... because the first law is not subject to such a restriction. It describes transitions between equilibrium states but is not concerned with the intermediate states. The system does not have to pass through only equilibrium states.

Example 3.3

Polishing a Fitting

A machinist polishes a 0.50-kg copper fitting with a piece of emery cloth for 2.0 min. He moves the cloth across the fitting at a constant speed of 1.0 m/s by applying a force of 20 N, tangent to the surface of the fitting. (a) What is the total work done on the fitting by the machinist? (b) What is the increase in the internal energy of the fitting? Assume that the change in the internal energy of the cloth is negligible and that no heat is exchanged between the fitting and its environment. (c) What is the increase in the temperature of the fitting?

Example 3.5

Isothermal Expansion of an Ideal Gas

Heat is added to 1 mol of an ideal monatomic gas confined to a cylinder with a movable piston at one end. The gas expands quasi-statically at a constant temperature of 300 K until its volume increases from V to $3V$. (a) What is the change in internal energy of the gas? (b) How much work does the gas do? (c) How much heat is added to the gas?

A **quasi-static process** refers to an idealized or imagined process where the change in state is made infinitesimally slowly so that at each instant, the system can be assumed to be at a thermodynamic equilibrium with itself and with the environment.

Quasi-static processes are done slowly enough that the system remains at thermodynamic equilibrium at each instant, **despite the fact that the system changes over time**. The thermodynamic equilibrium of the system is necessary for the system to have well-defined values of macroscopic properties such as the temperature and the pressure of the system at each instant of the process. Therefore, quasi-static processes can be shown as well-defined paths in state space of the system.

... the process is conducted in infinitesimal steps **such that these departures from equilibrium can be made as brief and as small as we like**.

A **reversible process** is one that can be made to retrace its path by differential changes in the environment. Such a process must therefore also be quasi-static. Note, however, that **a quasi-static process is not necessarily reversible**, since there may be dissipative forces involved.

Example 3.7

Compression of an Ideal Gas in an Automobile Engine

Gasoline vapor is injected into the cylinder of an automobile engine when the piston is in its expanded position. The temperature, pressure, and volume of the resulting gas-air mixture are $20\text{ }^{\circ}\text{C}$, $1.00 \times 10^5\text{ N/m}^2$, and 240 cm^3 , respectively. The mixture is then compressed adiabatically to a volume of 40 cm^3 . Note that in the actual operation of an automobile engine, the compression is not quasi-static, although we are making that assumption here. (a) What are the pressure and temperature of the mixture after the compression? (b) How much work is done by the mixture during the compression?

Questions

